

Reviewer Pack — Symmetric Group Character Discrepancy Lower Bound for Matrix...

Entry b84510964185 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Symmetric Group Character Discrepancy Lower Bound for Matrix Rigidity

Entry ID: b84510964185 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-20 09:04:13 UTC

1. Conjecture

Field A (mathematical branch): Representation Theory of Symmetric Groups **Field B** (complexity object): Communication Complexity of Disjointness

Statement:

For an $n \times n$ matrix M with entries ± 1 , let $\chi_\lambda(M)$ denote the sum of absolute values of its irreducible character projections under the symmetric group S_n . Then $\chi_\lambda(M) \geq \Omega(n^{\{1.5\}})$ for all λ with $|\lambda| \leq n^{\{1.5\}}$, and $\chi_\lambda(\text{perm}_n) > \chi_\lambda(\text{det}_m)$ for all $m < n^{\{1.5\}}$.

Rationale (proposer's reasoning):

Symmetric group characters capture combinatorial symmetries of matrices, while communication complexity lower bounds for disjointness rely on matrix rigidity. The character sum $\chi_\lambda(M)$ measures how 'rigid' a matrix is under permutation symmetries, potentially exposing structural gaps between permanent and determinant matrices.

Taxonomy category: COMM_COMPLEXITY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): d7f06c75617dc98d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.95	SAFE	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction
from itertools import combinations_with_replacement, chain

def partitions(n):
    if n == 0:
        yield []
    else:
        for p in partitions(n - 1):
            for i in range(len(p) + 1):
                yield p[:i] + [p[i-1] + 1] + p[i+1:] if i > 0 else [n]

def chi_lambda(M, lambda_):
    n = len(M)
    sign = Fraction(1)
    for mu in partitions(n):
        if len(mu) == len(lambda_) and sum(mu) == sum(lambda_):
            product = 1
            for i in range(n):
                for j in range(i + 1, n):
                    if M[i][j] != M[j][i]:
                        return None
            product *= Fraction(1 - (-1)**(mu[i] + mu[j]))
        sign *= product
    return abs(sign)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = 40
    m = 20
    M = [[random.choice([-1, 1]) for _ in range(n)] for _ in range(n)]

    chi_perm_n = chi_lambda(M, [n])
    if chi_perm_n is None:
        return {
            "metric_name": "chi_lambda",
            "metric_value": None,
```

```

        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

chi_det_m = chi_lambda(M, [m])
if chi_det_m is None:
    return {
        "metric_name": "chi_lambda",
        "metric_value": None,
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

metric_value = chi_perm_n - chi_det_m
conjecture_holds = metric_value > 0
counterexample = "" if conjecture_holds else f"chi_det_{m} >= chi_perm_{n}"

return {
    "metric_name": "chi_lambda",
    "metric_value": metric_value,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results if r["metric_value"] is not None) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        for r in results:
            if not r["conjecture_holds"]:
                counterexample = r["counterexample"]
                first_failing_seed = seed
                break

        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
TRIAL: {'metric_name': 'chi_lambda', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'chi_lambda', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'chi_lambda', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'chi_lambda', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aa945edf.py", line 83, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aa945edf.py", line 45, in run_trial
    chi_perm_n = chi_lambda(M, [n])
                  ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aa945edf.py", line 35, in chi_lambda
    product *= Fraction(1 - (-1)**(mu[i] + mu[j]))
                               ~~~~~
```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed with IndexError before producing valid results; conjecture validity cannot be assessed. | next: Fix the IndexError in chi_lambda function by validating mu list bounds and re-run tests with multiple seeds

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	80906
2	preregistration	ollama_remote	qwen3:8b	0	0	31723
3	novelty	ollama_remote	qwen3:8b	0	0	27376
4	novelty	ollama_remote	qwen3:8b	0	0	20130
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14389
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11642
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8752
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11076
9	judge	ollama_remote	qwen3:8b	0	0	23618

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 229612 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/b84510964185.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/b84510964185.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/b84510964185.tar.gz (if generated)